Solution / Buns

2110201: COMP ENG MATH (2023/2) Quiz 3

Name: Solution Student ID: 0123456789 Entry Number: 0

1. (4 points) Find the determinant of the matrix $\begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -3 & 0 & 2 \\ -2 & -1 & 2 & -3 \\ 3 & 0 & 1 & x \end{bmatrix}$. where x represents the 8th digit in your student ID. For example, if your ID is 6531234**5**21, then x = 5 for your matrix.

[**Hint:** The answer is an integer.]

Note: The answer is the same for all x! It's just a trick!

$$\begin{vmatrix} 1 & 2 & -1 & 3 \\ 0 & -3 & 0 & 2 \\ -2 & -1 & 2 & -3 \\ 3 & 0 & 1 & x \end{vmatrix} \xrightarrow[R_4-3R_1]{R_3+2R_1} \begin{vmatrix} 1 & 2 & -1 & 3 \\ 0 & -3 & 0 & 2 \\ 0 & 3 & 0 & 3 \\ 0 & -6 & 4 & x-9 \end{vmatrix} \xrightarrow[R_4-2R_2]{R_3+R_2} \begin{vmatrix} 1 & 2 & -1 & 3 \\ 0 & -3 & 0 & 2 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 4 & x-13 \end{vmatrix}$$

$$\xrightarrow{R_3 \leftrightarrow R_4} (-1) \begin{vmatrix} 1 & 2 & -1 & 3 \\ 0 & -3 & 0 & 2 \\ 0 & 0 & 4 & x-13 \\ 0 & 0 & 0 & 5 \end{vmatrix} = (-1) \times (1 \times (-3) \times 4 \times 5) = 60 \ \#$$

2. (4 points) Find the inverse (if any) of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \\ 0 & 6 & 9 \end{bmatrix}$. → Draw augmented matrix $[A \mid I] \Rightarrow [RRE(A) \mid A^{-1}]$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 1 & 1 & 2 & 0 & 1 & 0 \\ 0 & 6 & 9 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_2-R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -1 & -1 & -1 & 1 & 0 \\ 0 & 6 & 9 & 0 & 0 & 1 \end{array}\right] \xrightarrow{(-1)R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & -1 & 0 \\ 0 & 6 & 9 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3-6R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & -1 & 2 & 0 \\ 0 & 1 & 1 & 1 & -1 & 0 \\ 0 & 0 & 3 & -6 & 6 & 1 \end{array}\right] \xrightarrow{\frac{1}{3}R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & -1 & 2 & 0 \\ 0 & 1 & 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & -2 & 2 & \frac{1}{3} \end{array}\right]$$

$$\xrightarrow[R_2-R_3]{R_1-R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & -\frac{1}{3} \\ 0 & 1 & 0 & 3 & -3 & -\frac{1}{3} \\ 0 & 0 & 1 & -2 & 2 & \frac{1}{3} \end{array}\right]$$

I Ans

$\therefore$ Ans $\begin{bmatrix} 1 & 0 & -\frac{1}{3} \\ 3 & -3 & -\frac{1}{3} \\ -2 & 2 & \frac{1}{3} \end{bmatrix}$

3. (2 points, 0.5 points each) Given a matrix A with the properties in each sub-question. Is the matrix A invertible? (Answer **Yes** or **No** in the table provided.)

3.1	3.2	3.3	3.4
Yes	No	Yes	Yes

3.1. $\mathbf{A} = \mathbf{I}_4$ (I is always invertible, $A^{-1} = I$)

3.2. $\text{RRE(A)} = \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 5 \\ 0 & 0 & 0 \end{bmatrix}$ ($RRE(A) \neq I_n$; A is not invertible)

3.3. $\mathbf{A}_{3\times3}$ and there exists a matrix $\mathbf{B}$ such that $\mathbf{BA} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ ($BA = 3I_3 \rightarrow A^{-1} = \frac{1}{3}B$)

3.4. $\mathbf{A}^{2024} = \mathbf{I}_{2567}$ ($A^{-1} = A^{2023}$)